

"You Need to Know What I Can Do": Revealing Sighted People's Workplace Bias Toward the Visually Impaired and Exploring Intervention Through Serious Games

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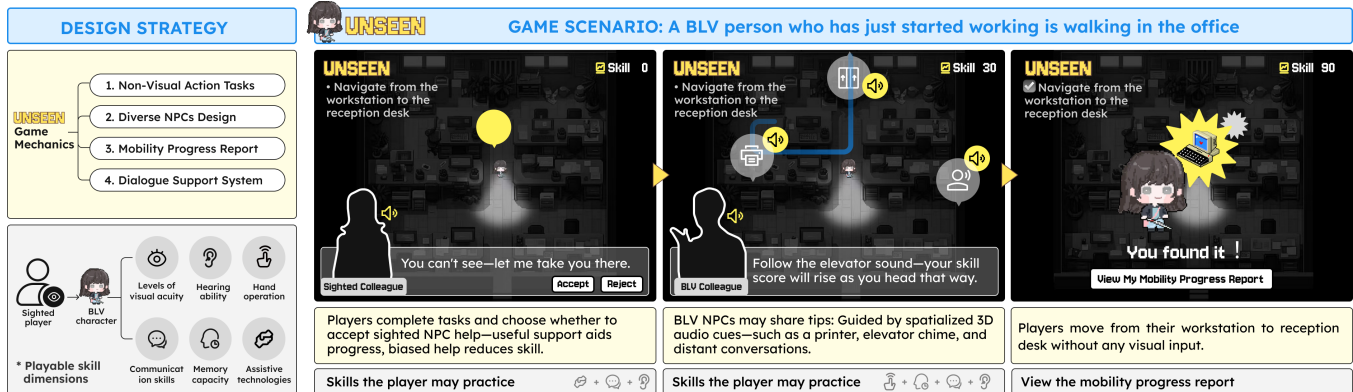


Figure 1: UNSEEN lets sighted players role-play as BLV individuals and personally experience BLV's abilities and challenges in a virtual workplace. In the "Workstation to Reception Desk" scenario: Step 1, players identify bias during a voice interaction and decide whether to accept help; Step 2, if help is rejected, they navigate using sound cues with optional BLV mentor guidance and subtle error feedback; Step 3, a "Mobility Progress Report" tracks performance, emphasizing that BLV mobility skills can be learned.

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CSCW Companion '25, Bergen, Norway
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ACM ISBN 979-8-4007-1480-1/25/10
<https://doi.org/10.1145/3715070.3749251>

Abstract

As workplace inclusion gains attention, expanding career opportunities for Blind and Low Vision (BLV) individuals is essential to their social participation and self-actualization. However, most research centers on assistive technologies and individual coping, overlooking societal-level discrimination—especially ableist bias from some sighted colleagues. Based on interviews with 11 BLV professionals and collaborative autoethnographies by 5 sighted researchers, this study identifies four unconscious ableist biases in

sighted people and shows how they limit BLV individuals' workplace inclusion. To counter their unintentional and persistent nature, we explore serious games as an intervention tool. By enabling embodied perspective-taking and trust-building, such games offer an engaging and empathetic way to shift perceptions. We present design strategies and a prototype framework, contributing a novel approach to supporting workplace inclusion through interactive, reflective experiences.

CCS Concepts

• **Social and professional topics** → **People with disabilities**;
 • **Human-centered computing** → **Empirical studies in HCI**;
Interaction design theory, concepts and paradigms.

Keywords

Visual impairment; Workplace exclusion; Ableist biases and perspective-taking; Serious games

ACM Reference Format:

Qixuan Xu, Ruolan Hu, Yilin Yuan, Yuxin Deng, Tianhui Xiong, Shuchang Xu, and Guanhong Liu. 2025. "You Need to Know What I Can Do": Revealing Sighted People's Workplace Bias Toward the Visually Impaired and Exploring Intervention Through Serious Games. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '25)*, October 18–22, 2025, Bergen, Norway. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3715070.3749251>

1 Introduction and Related Work

In contemporary workplaces, inclusive employment for blind and low vision (BLV) users remains a challenge. While some roles may require visual capabilities, a growing body of HCI research has addressed BLV users' access to technology, developing tools such as accessible coding environments [11, 15, 16], collaborative systems for mixed-vision teams [4, 6, 8, 17, 25], visual description support [30, 31], and enhanced assistive technologies [18, 20, 32, 33]. These efforts improve technical access and skill-building within the BLV community.

However, technical capability is only part of the problem. Many BLV individuals report being overlooked or excluded not because they cannot perform tasks, but because they are presumed unable to. This reflects a deeper social barrier: ableist bias—a form of prejudice that assumes lower competence based on disability status. Such bias imposes implicit "ceilings" on career progression [1, 19, 34], and even workplace exclusion [22, 34], often limiting BLV participation regardless of actual skills.

Existing research has documented how ableist bias channels BLV individuals into narrow occupational roles (e.g., massage, music), reinforcing low-expectation pathways [1, 14]. Even in policy contexts such as China's Disability Protection Law, practical barriers persist—for example, disqualifying candidates during medical exams based on vision standards [28]. Meanwhile, studies consistently show that BLV individuals often exceed societal expectations in learning and work contexts [1, 15, 16, 23, 29]. Yet ableist attitudes among some sighted colleagues continue to negatively impact their career development.

Efforts to reduce bias have largely focused on didactic education or accessibility training, but these approaches often fail to engage

participants or prompt meaningful reflection. More engaging, experiential interventions are needed.

Serious games offer an alternative by combining immersive interaction with emotional resonance. Prior work highlights their potential to: (1) lower psychological defensiveness through gamified framing [27]; (2) foster empathy via role-play and narrative [13, 26]; and (3) support theory-driven bias correction [9, 12]. These qualities make them well-suited to address unconscious ableist bias and reshape social cognition [2, 12, 24].

Yet, research on serious games for BLV inclusion remains limited. Most existing games target BLV skill training, while neglecting biases held by some sighted collaborators. And many designs rely heavily on "blindness simulation" rather than showcasing the adaptive capabilities of BLV users [10, 13].

This study addresses two research questions:

(1) How do unconscious ableist biases among some sighted individuals limit workplace inclusion for the BLV community?

(2) How can serious games be designed to address such bias and support more inclusive workplace practices?

Through interviews with 11 BLV professionals and collaborative autoethnographies, We identified four forms of ableist bias among sighted individuals and revealed how these biases contribute to the workplace exclusion of BLV individuals. In response, we propose a design framework for serious games that combines everyday workplace scenarios with psychological insights, offering a new path for systemic bias intervention.

Contributions of this work include:

(1) Revealing how ableist biases of some sighted individuals shape workplace exclusion for BLV individuals.

(2) Bridging a research gap by defining actionable design principles for serious games that target bias among sighted colleagues.

2 Methodology

2.1 Semi-structured Interview

We conducted semi-structured interviews with 11 individuals with visual impairments in two phases. The first phase involved exploratory interviews to identify key career-related challenges and refine the study's focus. The second phase involved in-depth interviews exploring employment status, transition motivations, decision-making, structural barriers, and job expectations. Demographic information was also collected (Appendix Table 1). All interviews were audio-recorded with consent and transcribed for analysis.

2.2 Collaborative Autoethnography

To better understand how sighted individuals perceive BLV experiences, we used collaborative autoethnography (CAE), combining ethnographic inquiry, personal writing, and reflection [21]. Five sighted researchers documented their interactions with the BLV community and reflected on their biases. Following Chang et al. (2016), we structured the process to include [7]: defining themes, clarifying positionalities, setting collaboration protocols, writing personal narratives, and jointly analyzing reflections.

2.3 Data Analysis

We thematically analyzed interview transcripts and CAE narratives [5]. Five researchers independently coded the data using mind maps.

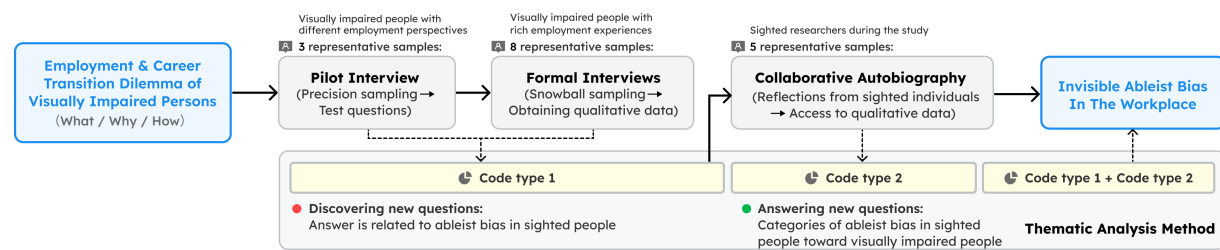


Figure 2: Research approaches and methods

Interviews were double-coded, and CAE narratives reviewed by separate team members. Discrepancies were discussed in group sessions to form a shared coding framework. The analysis revealed recurring patterns: BLV individuals can find jobs but struggle to advance, workplace exclusion is subtle, and sighted colleagues often hold unconscious ableist bias. These were organized into categories and subthemes, resulting in a core themes: the often-unconscious ableist assumptions held by sighted people, which shape and constrain workplace opportunities for BLV individuals.

3 Results

While BLV limit access to some professions, their career paths are also constrained by ableist bias. As P7 noted: *"You hear so many limiting voices—even family thinks massage is your only option. Eventually, you start to believe it."* (P6, P10). These biases are reinforced by structural barriers. P8 recalled: *"Although blind students shared the same high school curriculum as sighted peers, only massage and music were offered as college majors."* Certification systems also lack accommodations. P11 said: *"I wanted to take the public nutritionist exam but wasn't eligible."* P4 added: *"Some peers topped civil service exams but were disqualified due to vision requirements."*

P10 reflected: *"If even 10% of people dropped their bias, our job options would expand significantly."* These conversations also led our sighted research team to confront their own assumptions. Through reflective writing, we saw how deeply embedded—and often unconscious—these biases are. Based on interviews with 11 BLV participants (P1–P11) and collaborative autoethnography (CAE) by five sighted interviewers (I1–I5), our analysis revealed a central concern: the often-unconscious ableist assumptions held by sighted people, which shape and constrain workplace opportunities for BLV individuals. We categorize four common forms of bias that, while often well-intentioned, are pervasive in the workplace and constrain BLV career development.

(1) Logistical Focus Overlooks Competence. Sighted individuals often view BLV through a deficit lens, focusing on logistical challenges rather than workplace contributions. This framing creates invisible limits on what BLV are perceived to be capable of. As P10 noted: *"They focus on small details: how you find a meeting room, get food, or commute—and overlook your actual abilities."* P11 added: *"We're fully capable, but stereotypes still make people doubt us."*

This bias also surfaced in the CAE. I5 asked: *"Can blind people manage their appearance? How do they receive education? What materials do they use?"* I2 reflected: *"We were just curious how she helped a child search the trash bin—but too many questions seemed to*

upset her." When sighted people fixate on daily logistics rather than professional skills, they undermine perceptions of BLV as equal colleagues.

(2) Individual Cases Misrepresent the Group. BLV individuals vary widely in vision and careers—our interviews with 11 BLV people revealed 19 different jobs. Despite this diversity, sighted individuals often focus on limited, common roles like massage, reinforcing the stereotype that "blind people can only do massage." Meanwhile, a few online success stories have created the impression that "BLV no longer face career barriers", yet as P5 observed: *"I often see blind people with impressive jobs online, but most around me still do massage work"*, highlighting the gap between perception and reality.

This overgeneralization was evident in the CAE. I4 shared: *"After interviewing P1, I thought all blind people had good jobs—until I learned about the real challenges."* I5 added: *"I was shocked to realize how many career options blind people actually have—but then I asked myself, why was I so surprised?"* Such thinking reduces systemic barriers to individual exceptions, mistaking outliers for the norm and obscuring the broader struggles BLV face in professional life.

(3) Disability-Centric Views Limit BLV's Autonomy. Although BLV individuals do experience visual impairments, sighted people often exaggerate these limitations, leading to distorted evaluations of their workplace capabilities. As P10 noted: *"People assume your ability is low—so if you exceed it, they think it's impossible."* This forces BLV individuals to constantly prove themselves. P1 recalled: *"I had to open my laptop in the interview to show I could use it."* P7 added: *"I'm fully capable, but people assume otherwise."* Even after hiring, "well-intentioned" substitution persists. P10 shared: *"My boss gave me an assistant to 'improve efficiency', but I didn't need one."* Such help replaces autonomy instead of enabling it, deepening marginalization.

This bias also emerged in the CAE. As I1 remarked: *"Some of them are really good at programming—I just can't imagine how they understand code."* Such expressions of disbelief effectively deny BLV individuals their rightful identity as competent professionals.

(4) Caution in Interaction Hinders Inclusion. Although BLV individuals are often viewed as marginalized, excessive politeness rooted in fear of offense can feel alienating. As P4 put it: *"You don't need to use honorifics—I don't need that kind of sympathy."* P9 added: *"We can work, have hobbies—we're just ordinary people who happen to be blind."*

Workplaces are social as well as professional spaces. When sighted colleagues overcompensate with kindness, "social protection" often becomes "social distancing", limiting BLV participation.

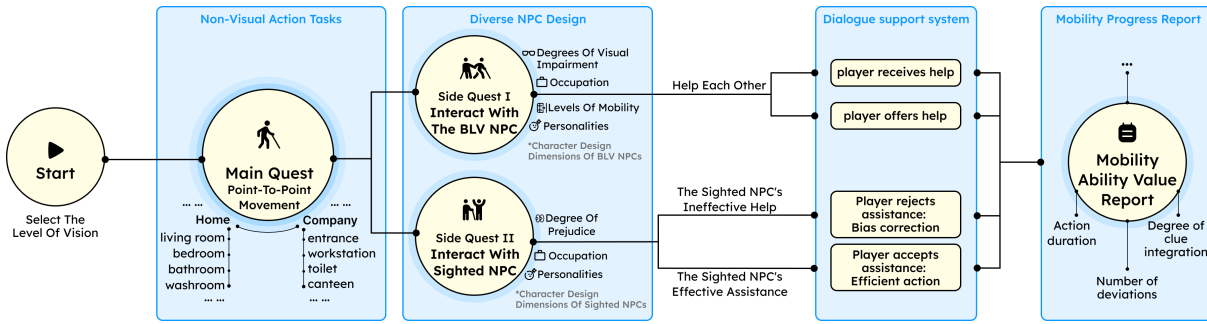


Figure 3: The game mechanism framework of <UNSEEN>

I1 noted: "I avoided asking about their work to not offend them, so I just assumed they could do it—without knowing how." I4 reflected: "I found myself offering extra positive feedback and being overly cautious in conversation. In hindsight, it felt like condescending care." Such well-intentioned but anxious behaviors deepen social divides and obscure BLV's role as equal collaborators.

4 Design Consideration

Our analysis shows that sighted individuals' ableist biases affect BLV individuals' access to jobs, interviews, and promotions, causing many capable BLV to miss out on ideal roles and respect. Therefore, addressing these biases is central to our design consideration.

4.1 Design Strategy

CAE showed that directly calling sighted people's attitudes "bias" often harms their self-esteem and triggers defensiveness [3], reinforcing stereotypes. As I5 noted: "A week ago, I'd have denied any bias and defended myself." Thus, gentle, low-threat interventions are essential. Serious games, with their embodied, contextual nature, evoke empathy and reshape cognition, making them ideal for addressing unconscious bias. We propose a three-step approach—make bias visible, trigger reflection, and build trust—guiding sighted players to reframe their views of BLV, leading to three design strategies.

(1) Reframing Capability Perceptions. To challenge sighted individuals' one-dimensional views of BLV and their abilities, the game introduces diverse avatars and NPCs that reflect the heterogeneity of the BLV community. Players rely on non-visual senses to complete basic tasks and navigate realistic workplace scenarios. Through embodied interaction and context-rich situations, the game encourages sighted users to reassess their assumptions about BLV cognition and capabilities.

(2) Unlocking Workplace Potential. To improve sighted individuals' understanding of BLV colleagues' problem-solving skills and growth potential, the game presents layered non-visual tasks that require strategic adaptation and repeated trials. Through feedback and progression systems, players experience a "novice-to-expert" learning path that highlights BLV individuals' professional capacity and adaptability.

(3) Bridging Social Distance. To address sighted individuals' tendency to withdraw from social interactions with BLV individuals

for fear of causing offense, the game simulates scenarios involving social misunderstandings, resolution, and eventual empathy-building. Players learn to respond through active listening and dialogue, followed by guided reflection to uncover their own blind spots and improve emotional readiness for real-world interactions.

4.2 Game Mechanics

Based on our design strategies, we propose UNSEEN, a serious game prototype set in the daily work life of BLV individuals. Players navigate workplace scenarios first-person, practicing route memorization, handling unexpected events, and completing non-visual tasks. This embodied experience reveals BLV mobility skills and workplace ableist biases. We introduce four mechanics using interactive, contextual tasks to raise bias awareness and deepen understanding of BLV abilities.

(1) Non-Visual Action Tasks. To counter one-dimensional views focusing on "invisibility" or "daily logistics," <UNSEEN> challenges players with non-visual tasks like path planning, sound localization, and rhythm control, simulating how BLV rely on other senses.

(2) Diverse NPCs Design. <UNSEEN> features blind NPCs with varied vision levels, mobility, and interaction styles, helping players appreciate the diversity within the BLV community.

(3) Dialogue Support System. Overworry about offending BLV can cause hesitant, vague communication and social withdrawal. <UNSEEN> simulates this with delayed or interrupted NPC responses. When players seek info as BLV characters, sighted NPCs show hesitation (e.g., "Uh... is this okay?"). The system prompts players to notice this and respond, encouraging dialogue.

(4) Mobility Progress Report. Sighted people often hold a static view of BLV mobility levels. The game uses repeatable navigation tasks to simulate skill growth in novice BLV users, tracking and visualizing progress to show that mobility skills develop through practice, not innate ability.

4.3 Design Reflection and Future Work

As more socially responsible companies hire employees with BLV, the serious game UNSEEN offers a promising tool for workplace inclusion. Suitable for onboarding, diversity workshops, and vocational training, it lets sighted players experience the skills and strategies of visually impaired workers firsthand, effectively challenging stereotypes.

In future work, we will conduct preliminary playtests and usability evaluations of the *UNSEEN* prototype to assess our design strategies. Sighted participants with workplace experience will test key scenarios, and we aim to involve BLV individuals in co-creation to enhance authenticity and diversity. Evaluation will cover task completion, cognitive load, emotional engagement, and empathy, using SUS and a simplified bias-reflection scale. Feedback will inform refinements to game mechanics, adaptive difficulty, and personalized feedback. We also plan to strengthen social interaction design to foster empathy and reduce misunderstanding-based avoidance.

5 Conclusion

This study identified ableist biases among some sighted individuals and introduced a serious game intervention. Drawing on BLV experiences and reflective autoethnography, we developed a dual-perspective approach. The game recreates workplace bias scenarios through interactive, non-visual tasks to surface implicit attitudes. Though rooted in China, patterns like underestimation and social distancing hold global relevance. Our embodied design promotes perspective-taking and offers an empathetic, cross-cultural path to reduce bias and foster inclusion.

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A Appendix

Table 1: Demographic and Background Information of Participants

No.	Gender	Types of Visual Impairment	Educational Level	Identity	Region
P1	F	Blind with a sense of light	Bachelor degree	Accessibility Testing Engineer	Beijing
P2	M	Acquired total blindness	Associate degree	Massage therapist	Beijing
P3	M	Congenital low vision	Bachelor degree	Student	Australia
P4	F	Blind with a sense of light	Bachelor degree	Student	Jilin
P5	M	Congenital blindness	Associate degree	Accessibility Testing Engineer	Zhejiang
P6	F	Acquired low vision	High school diploma	Stay-at-home mom	Shanghai
P7	F	Acquired low vision	Bachelor degree	Data annotator	Shanghai
P8	M	Blind with no sense of light	Associate degree	Data annotator	Shanghai
P9	F	Acquired total blindness	Bachelor degree	Bank clerk	Shanghai
P10	M	Acquired total blindness	Incomplete bachelor's degree	Accessibility Testing Engineer	Beijing
P11	F	Acquired total blindness	High school diploma	Data annotator	Shanghai
I1	F	Normal vision	Master's degree	Student	Shanghai
I2	F	Normal vision	Master's degree	Student	Shanghai
I3	F	Normal vision	Master's degree	Student	Shanghai
I4	F	Normal vision	Master's degree	Student	Shanghai
I5	F	Normal vision	Master's degree	Student	Shanghai